

System Architecture Design

High-Level Design (HLD) - system-oda big picture

Architecture Design na yenna?

System-oda main parts (components) yenna, avanga epdi connect aagudhu, yenna technology use pannuvom nu **overall structure** decide pannradhu.

Architecture Styles

Style	Meaning	Yeppo use pannalam
Layered (3-tier)	UI, Business logic, Database-nu layers.	Simple web / college projects.
Client-Server	Client request anuppum, server respond pannum.	Most web and mobile apps.
MVC	Model, View, Controller-nu pirikkum.	Web frameworks (Django, Spring).
Microservices	Chinna independent services.	Periya, scalable systems.
Monolithic	Ellam oru application-la.	Small projects, quick start.

Main Steps

- **Step 1:** Requirements-a padinga (functional + non-functional).
- **Step 2:** Major components identify pannunga (Frontend, Backend, Database, APIs).
- **Step 3:** Architecture style select pannunga.
- **Step 4:** Tech stack decide pannunga.
- **Step 5:** Architecture diagram varainga.
- **Step 6:** Risks-um trade-offs-um ezhudhunga.

Tech Stack Decision Table

Layer	Option	Yen select pannom
Frontend		
Backend		
Database		
Hosting		

Design Quality Points

- Scalability - users adhigam aanaalum work aaganum.
- Security - data-va protect pannanum.

- Maintainability - later maathradhu easy-a irukkanum.
- Performance - speed.

Tip: Architecture diagram-la box + arrow mattum pothum. Simple-a vaikkonga.